ModelIt!

Student Activity Pack

Designing a Cancer Drug That Actually Works

Multi-Scale Pharmacokinetic Modeling from Molecule to Patient

Grade 09L3 | NGSS HS-LS1-4, HS-ETS1-2 | Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Why do you think cancer treatment often makes patients extremely sick — sometimes sicker than the cancer itself?**

2. **What do you think determines whether a drug dose is 'just right' versus 'too much' or 'too little'?**

3. **Draw a diagram showing how you think a drug travels through the body from the moment it's administered to the moment it reaches a tumor.**

4. **What is a therapeutic window and why might it matter for cancer treatment?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Drug Dosage | Absorption Rate | Blood Concentration | Tumor Uptake | Healthy Cell Damage | Metabolism Rate | Kidney Clearance | Therapeutic Window | Side Effect Severity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Standard Dosage

Recommended clinical dose, normal metabolism

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: High-Dose Aggressive

Double the standard dose

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Metronomic Optimization

Half-dose given twice as frequently

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Tumor Blood Supply — The density and permeability of blood vessels feeding the tumor, which determines how much circulating drug can actually reach cancer cells — tumors with poor blood supply may be resistant even at high doses

Drug Resistance Factor — The probability that cancer cells develop molecular mechanisms to pump the drug out or deactivate it, reducing effectiveness over multiple treatment cycles

Patient Liver Function — The individual patient's liver enzyme activity, which determines how quickly the drug is metabolized — liver impairment can cause dangerous drug accumulation

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do most cancer drugs fail in clinical trials — and how can computational modeling help us design ones that actually work? Use evidence from your simulation to support your answers.

1. **How does your multi-scale model demonstrate the challenge of connecting molecular drug behavior to patient-level outcomes?**

2. **Why does the same drug dose produce different outcomes in different patients, according to your model?**

3. **What trade-offs did you discover between maximizing Tumor Uptake and minimizing Side Effect Severity?**

4. **How could your model be used to personalize cancer treatment for individual patients?**

5. **What are the limitations of modeling a 9-component pharmacokinetic system compared to the full complexity of cancer biology?**

8 STEM Challenge: Design an Adaptive Chemotherapy Dosing Algorithm

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YOUR ENGINEERING MISSION

Design a computational algorithm that adjusts chemotherapy dosing in real time based on patient biomarker data, optimizing tumor destruction while minimizing side effects.

THE SCENARIO

A pharmaceutical company is developing a next-generation adaptive dosing system for cancer treatment. Instead of giving every patient the same dose on a fixed schedule, the system would use real-time blood biomarker data to adjust doses automatically. Your team has been hired to design the algorithm that determines when and how much drug to administer.

Engineering Design Thinking

1. HOW FREQUENTLY SHOULD THE SYSTEM SAMPLE BLOOD BIOMARKERS — CONTINUOUSLY OR AT INTERVALS?
2. WHAT MACHINE LEARNING APPROACH COULD IMPROVE DOSING PREDICTIONS OVER MULTIPLE TREATMENT CYCLES?
3. HOW WOULD YOU VALIDATE YOUR ALGORITHM BEFORE TESTING IT ON REAL PATIENTS?
4. WHAT ETHICAL CONSIDERATIONS ARISE FROM AUTOMATING MEDICAL DOSING DECISIONS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Pharmacokineticists and Computational Pharmacologists design drug dosing protocols using mathematical models and computer simulations. They work for pharmaceutical companies, the FDA, and academic medical centers, earning $90,000–$180,000/year. Clinical Pharmacologists who manage cancer drug protocols earn $150,000–$300,000/year.